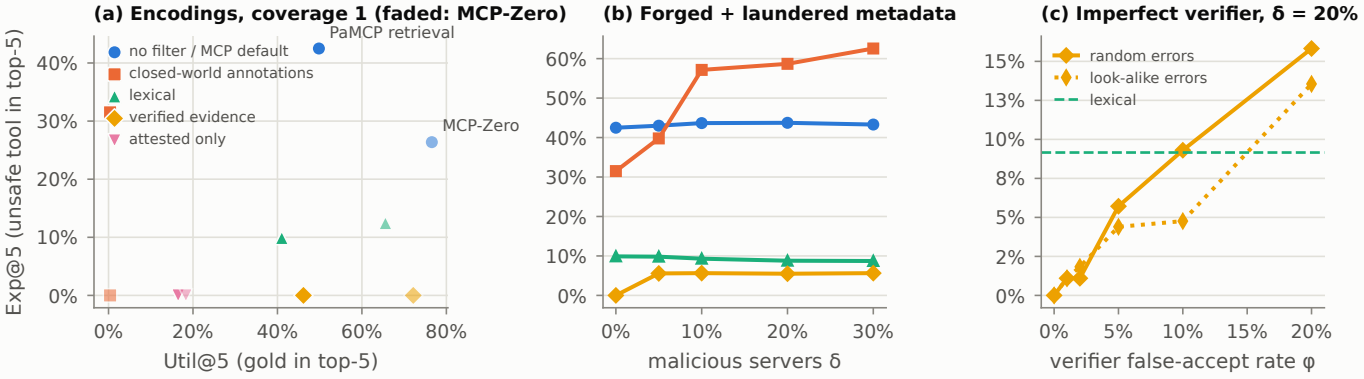
(a) Encodings, coverage 1 (faded: MCP-Zero)
no filter / MCP default
closed-world annotations
lexical
verified evidence
attested only
PaMCP retrieval
MCP-Zero
Exp@5 (unsafe tool in top-5)
Util@5 (gold in top-5)
0%
10%
20%
30%
40%
60%
80%
(b) Forged + laundered metadata
malicious servers δ
0%
10%
20%
30%
(c) Imperfect verifier, δ = 20%
random errors
look-alike errors
lexical
verifier false-accept rate φ
0%
2%
5%
8%
10%
13%
15%
20%